1.INTRODUCTION

1.1 overview

In recent years, technology has been playing an important role in connecting people and creating new career opportunities. One such development is the Integrated Smart Alumni Networking and Career Support Portal, which provides a common platform for students, alumni, and administrators to interact and share valuable information From building professional connections to providing career guidance, mentorship, and job opportunities, this portal helps students connect with experienced alumni and explore different career paths. It also allows alumni to share their professional experiences, post job opportunities, and guide students in their career development. From building professional connections to providing career guidance, mentorship, and job opportunities, this portal helps students connect with experienced alumni and explore different career paths. It also allows alumni to share their professional experiences, post job opportunities, and guide students in their career development. During this project, we will explore the major features of the portal, including alumni registration and login, profile management, alumni search, mentorship requests, job application tracking, dashboards, notifications, and administrative functionalities. This project aims to provide an organized digital platform that makes alumni networking easier and more effective. It helps students receive career guidance, professional advice, and mentorship from alumni working in different organizations and industries. The portal also supports communication between users and helps institutions maintain updated alumni information. It reduces communication gaps, minimizes manual work, and makes important information easily accessible.

1.2 Problem statement

In many educational institutions, communication between students and alumni is not properly organized. Alumni information, professional details, and career opportunities are often maintained through manual records, emails, or different social media platforms. This makes it difficult for students to identify and connect with suitable alumni for career guidance and mentorship. Students may face difficulties in finding job opportunities, receiving professional advice, and understanding current industry requirements. Similarly, alumni may not have a proper platform to share their experiences, provide mentorship, and communicate with current students. The absence of a centralized system also creates difficulties for administrators in maintaining alumni records and managing communication. The traditional alumni management system faces several challenges in maintaining communication between students, alumni, and administrators. Alumni information is often stored manually or across different platforms, making it difficult to search, update, and manage records. Students also face difficulties in finding experienced alumni for mentorship, accessing job opportunities, and tracking job applications.

1.3 Research Motivation

The main motivation behind developing the Integrated Smart Alumni Networking and Career Support Portal is to improve communication between students and alumni and provide better career support through a centralized digital platform.Many students face difficulties in finding proper career guidance, mentorship, and job opportunities. Alumni have valuable professional experience and industry knowledge, but there is often no organized platform to share this information with current students.

1.4 Research objectives:

The main objective of the Integrated Smart Alumni Networking and Career Support Portal is to develop a centralized platform that connects students, alumni, and administrators and supports career development.

The major objectives are:

- To develop a centralized platform connecting students and alumni.
- To improve communication between students and alumni.
- To provide alumni profile creation and management.
- To help students search for experienced alumni.
- To provide mentorship and career guidance.
- To share job opportunities and professional information.
- To support job application tracking.
- To provide dashboards and notifications.
- To simplify alumni data management for administrators.
- To strengthen relationships between the institution, students, and alumni.

1.5 significance:

The portal helps students receive career guidance, mentorship, and valuable industry knowledge from alumni. It also provides information about job opportunities and supports students in tracking their job applications. Alumni can share their professional experiences, guide students, and contribute to the development of the institution. The system reduces manual work and makes alumni information easier to maintain. It also improves communication through dashboards and notifications. Administrators can manage users and monitor portal activities efficiently. Overall, this project strengthens alumni relationships, promotes knowledge sharing, supports career development, and creates better opportunities for students to achieve their professional goals.

1.6 Advantages:

The Integrated Smart Alumni Networking and Career Support Portal provides several advantages to students, alumni, and educational institutions. It creates a centralized platform for communication, networking, and career development.

The main advantages of the proposed system are:

- Easy communication between students and alumni.
- Better professional networking opportunities.
- Access to career guidance and mentorship.
- Easy sharing of job opportunities.
- Improved job application tracking.
- Centralized alumni information management.
- Reduced manual work and time consumption.
- Quick updates through notifications.
- Better alumni participation in institutional activities.
- Improved career development and knowledge sharing.

1.7 Applications:

The Integrated Smart Alumni Networking and Career Support Portal can be used in educational institutions and organizations to improve alumni communication and career support.

The major applications of the system are:

- Educational Institutions: Maintains alumni information and strengthens student-alumni relationships.
- Career Guidance: Helps students receive professional advice from experienced alumni.
- Mentorship Programs: Connects students with alumni for career guidance and skill development.
- Job Opportunities: Allows alumni to share employment opportunities with students.
- Professional Networking: Helps students build connections with alumni working in different industries.
- Alumni Management: Supports institutions in maintaining alumni profiles and records.
- Placement Support: Helps students explore job opportunities and track applications.
- Institutional Development: Encourages alumni participation in college activities and knowledge-sharing programs.

2.LITERATURE SURVEY

Cinco and Dalan [1] developed a web-based Alumni Network and Database Information System. Their study focused on maintaining alumni information and improving communication between graduates.

Batistic and Tymon [2] studied networking behaviour and graduate employability. They found that professional networking helps students access useful information, resources, and employment opportunities.

English et al. [3] examined the importance of professional networks in graduate employability. Their research highlighted the role of alumni connections and social media in developing professional relationships.

Nabi et al. [4] reviewed mentoring in higher education and identified its importance in student career development, career choices, and professional growth.

Bolton-King [5] studied student mentoring and employability potential. The research highlighted that mentoring can improve confidence, self-efficacy, and professional skills.

Jackson et al. [6] examined student employability-building activities and highlighted the importance of mentoring, career counselling, and practical learning in preparing students for employment.

Trillano et al. [7] developed an Alumni Tracer and Management System to improve alumni engagement, graduate tracking, and data management in educational institutions.

Sawase et al. [8] presented an Alumni Connect System that provides networking, mentorship, communication, and job-sharing facilities for students and alumni.

Fermo et al. [9] developed an alumni tracking system that helps institutions monitor graduate employment information and maintain alumni records.

Malarvizhi et al. [10] proposed a digital platform for centralized alumni data management and engagement. Their research focused on improving alumni communication.

Shelar and Tawde [11] discussed a digital platform for alumni data management, mentorship programmes, and institutional engagement.

Palaskar et al. [12] presented a Student-Alumni Interactive Platform for Career Guidance and Networking to support communication between students, alumni, and administrators. Prasad et al. [13] proposed a digital platform for alumni engagement and networking with authentication, profile management, job opportunities, and event management.

Pawar et al. [14] proposed an AI-driven alumni-student collaboration system to improve alumni communication, career guidance, and mentorship opportunities.
Garvilles et al. [15] developed an Alumni Information Management System to centralize alumni data, improve networking, and support career development.
Cresencio [16] developed an Alumni Engagement Portal System for tracer studies. The research focused on improving alumni participation and replacing outdated manual processes.
The study Developing Professional Networks [17] highlighted that professional connections developed during university education can improve graduate employability and career opportunities.
Ahmed and Osman [18] reviewed the relationship between mentorship and graduate employability. Their study highlighted the importance of mentoring in preparing students for professional careers.
The study Career Resources and Securing Quality Work [19] examined how career resources, social support, and professional networks help graduates move into employment.
The study Student Employability-Building Activities [20] examined how career development activities contribute to graduate outcomes and employment preparation.
The study Networking Behaviour and Graduate Employability [21] explained how networking helps students access professional information and resources that support career development.
The study Alumni Tracking System for SEAIT [22] examined the use of human-computer interaction principles to improve alumni record management and institutional engagement.
Rejimole and Rajeswari [23] proposed an Alumni Association Management System that supports alumni registration, profile management, job postings, event updates, and communication.

3.EXISTING SYSTEM

3.1 overview

The Integrated Smart Alumni Networking and Career Support Portal is a web-based platform developed to connect students, alumni, and administrators through a centralized system. It provides an easy way to maintain alumni information, communicate with former students, and support students in their career development.The portal includes various features such as user registration and login, forgot password, alumni profile creation, alumni search, mentorship requests, job opportunities, job application tracking, dashboards, and notifications. Alumni can share their professional experiences, provide career guidance, and help students explore employment opportunities**.**

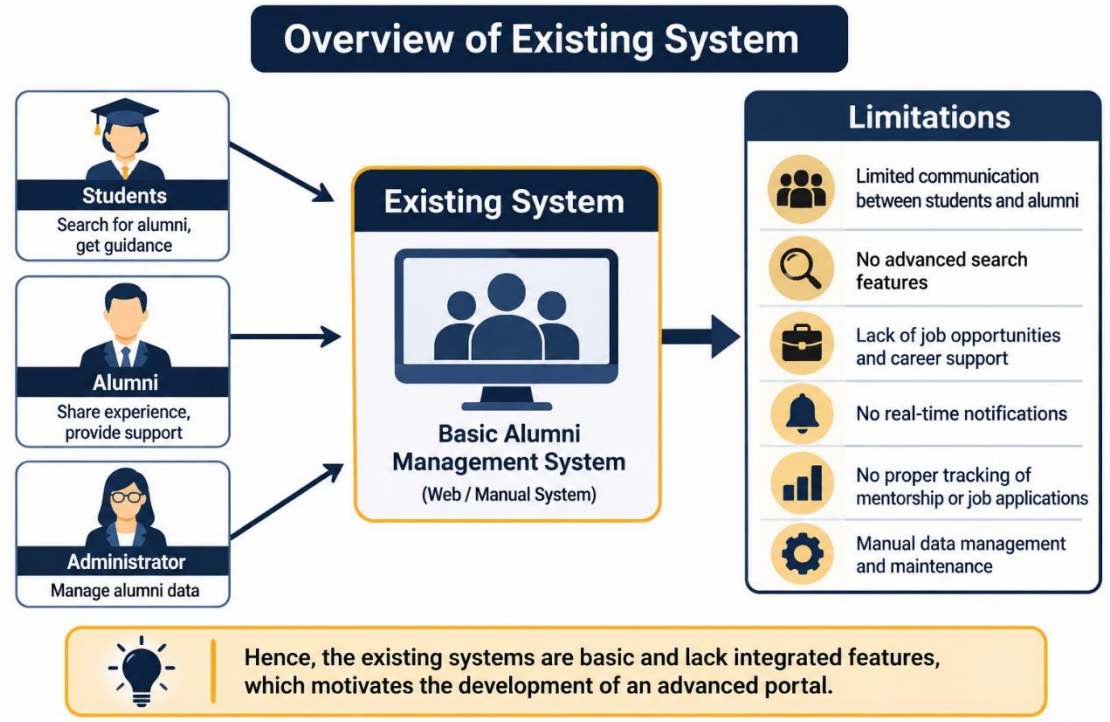


Fig 3.1 Overview of Existing System

3.2 Limitations of Traditional system

1. Manual Record Maintenance: Alumni details are maintained manually, making it difficult to update and manage information.

2. Limited Communication: Students and alumni face difficulties in maintaining regular communication

3. Lack of Mentorship: Students may not easily connect with experienced alumni for career guidance.

4. No Centralized Platform: Alumni information, job updates, and student details are scattered across different platforms.

5. Difficult Job Tracking: Students cannot easily track their job applications and recruitment progress.

6. Time-Consuming Process: Searching for alumni details and job opportunities requires more time and effort.

7. Limited Notifications: Students and alumni may miss important updates, events, and job notifications.

8. Poor Data Management: Maintaining accurate and updated alumni information becomes difficult.

9. Security Risks: Manual records may be exposed to unauthorized access.

10. Lack of Centralized Platform: Alumni information, mentorship, and job details are scattered across different sources.

4.PROPOSED SYSTEM

4.1 Overview:

The proposed Integrated Smart Alumni Networking and Career Support Portal is a web-based application designed to overcome the limitations of traditional alumni management systems. It provides a centralized platform for students, alumni, and administrators to communicate, share information, and support career development.Unlike traditional systems, this portal makes alumni information easily accessible and provides better communication between students and experienced alumni. It helps students build professional connections, gain industry knowledge, receive career guidance, and explore employment opportunities. Alumni can also contribute to the institution by sharing their experiences, providing mentorship, and posting job openings.

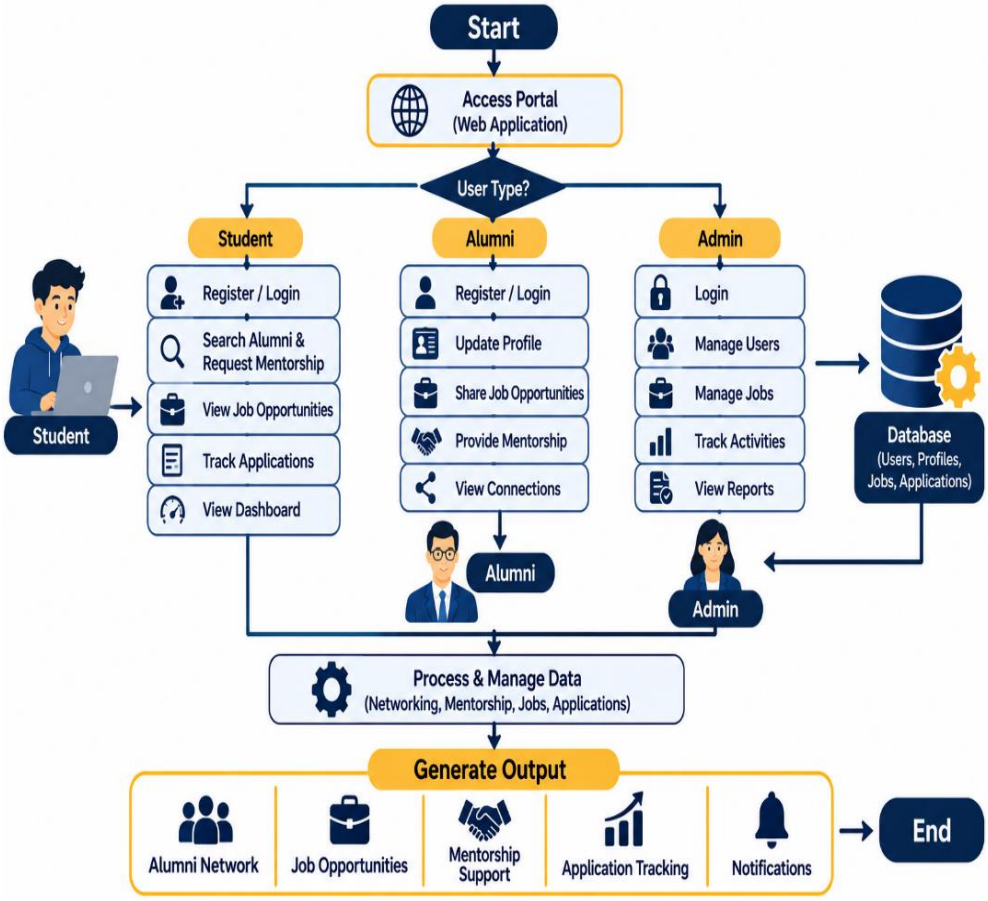


Fig 4.1 Overview of Proposed system

The system provides the following features:

1. User Registration: Allows students and alumni to create new accounts by entering their personal details.

2. User Login: Provides secure access to the portal using registered email IDs and passwords.

3. Forgot Password: Helps users recover their accounts when they forget their passwords.

4. Alumni Profile Management: Allows alumni to create, update, and manage their personal and professional profiles.

5. Alumni Search: Enables students to search for alumni based on name, department, graduation year, and professional details.

6. Mentorship Requests: Allows students to send mentorship requests to experienced alumni for career guidance.

7. Job Opportunities: Enables alumni to share job openings and employment opportunities with students.

8. Job Application Tracking: Helps students monitor their job application status and progress.

9. Student Dashboard: Displays student profiles, available jobs, mentorship activities, and application updates.

10. Alumni Dashboard: Allows alumni to manage profiles, view mentorship requests, and share job opportunities.

11. Notifications: Provides updates about job postings, mentorship requests, application status, and important activities.

12. Admin Functionality: Allows administrators to manage users, monitor activities, verify information, and maintain the portal.

4.2 workflow:

The workflow explains the sequence of steps involved in using the Integrated Smart Alumni Networking and Career Support Portal. It describes how users access the system, perform different operations, and receive results.

Step-by-step workflow:

1. Portal Access: Users open the portal through a web browser.

2. Registration: New students and alumni create accounts by entering the required details.

3. Login: Registered users enter their email ID and password to access the system.

4. Authentication: The system verifies the login credentials.

5. Role Identification: The system identifies the user as a student, alumni, or administrator.

6. Dashboard: Users are redirected to their respective dashboards.

7. Profile Management: Users can view and update their profile information.

8. Alumni Search: Students search for alumni using available search options.

9. Mentorship Process: Students send mentorship requests, and alumni can accept or reject them.

10. Job Process: Alumni share job opportunities, and students can apply for suitable jobs.

11. Application Tracking: Students check their job application status and receive notifications.

12. Admin Process: Administrators manage accounts, verify information, and monitor system activities.

13. Logout: Users log out after completing their activities.

4.3 react,js:

React.js is a popular JavaScript library used for building interactive and dynamic user interfaces. In our project, Integrated Smart Alumni Networking and Career Support Portal, React.js is used to develop the frontend of the application and provide a smooth user experience.It helps in creating different pages such as user registration, login, dashboards, alumni profiles, alumni search, mentorship requests, job opportunities, and admin management. React.js uses a component-based architecture, which allows us to divide the application into small, reusable components and makes development and maintenance easier.React.js also uses a Virtual DOM to update only the required parts of the user interface, improving application performance. It supports smooth navigation and provides an organized structure for managing different sections of the portal.Overall, React.js plays an important role in developing a responsive.

4.4: Flask framework:

Flask is a lightweight Python web framework used for developing web applications and backend services. It provides a simple and flexible environment for handling server-side operations.In our Integrated Smart Alumni Networking and Career Support Portal, Flask is used to handle backend operations such as user registration, login authentication, profile management, mentorship requests, and job-related activities. It also helps connect the frontend application with the database through APIs.Flask supports URL routing, request handling, and database integration, making it suitable for developing lightweight and efficient web applications.Overall, Flask helps manage backend functionality and provides communication between the React.js frontend and the database.Important: Include this only if you actually used Flask. If you tell me your backend technology, I can give you the exact framework description.

4.5 SQL:

SQL (Structured Query Language) is a standard language used to store, manage, retrieve, and manipulate data in relational databases. It helps users perform operations such as inserting, updating, deleting, and retrieving records from database tables.In our Integrated Smart Alumni Networking and Career Support Portal, SQL is used for managing important information such as student details, alumni profiles, login credentials, mentorship requests, job opportunities, and job application records.SQL provides different commands such as SELECT, INSERT, UPDATE, and DELETE to perform database operations. It also helps maintain organized data and retrieve information whenever required.Overall, SQL plays an important role in database management by providing an efficient way to store, organize, and access portal information.

5.SYSTEM ARCHITECTURE

The system architecture of the Integrated Smart Alumni Networking and Career Support Portal describes the overall structure and communication between different components of the application. It explains how the frontend, backend, and database work together to provide portal functionalities.The system follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Database Layer.

1. Presentation Layer (Frontend):

The Presentation Layer is the frontend part of the Integrated Smart Alumni Networking and Career Support Portal. It provides a user-friendly interface through which students, alumni, and administrators can interact with the system. React.js is used to develop the frontend interface. It allows users to register, log in, update profiles, search for alumni, send mentorship requests, explore job opportunities, track job applications, and receive notifications. Alumni can connect with students and share their career experiences, while administrators can access and manage portal activities. This layer sends user requests to the backend and displays the processed information to the users.

2. Application Layer (Backend):

The Application Layer is the backend part of the Integrated Smart Alumni Networking and Career Support Portal. It handles the business logic and processes requests received from the frontend. Node.js and Express.js are used to develop the backend. It manages user registration and login, alumni profile management, alumni searching, mentorship requests, job opportunities, job application tracking, notifications, and admin functionalities. It communicates with the database to store, retrieve, and update the required information. After processing the requests, it sends the response back to the frontend for displaying the results to users.

3. Database Layer:

The Database Layer is responsible for storing and managing all the information related to the Integrated Smart Alumni Networking and Career Support Portal. MySQL is used as the database to store student details, alumni profiles, administrator information, login credentials, mentorship requests, job opportunities, job application records, and notifications. It allows the backend to insert, retrieve, update, and delete information whenever required. It ensures that user information is organized and available whenever students, alumni, or administrators access the portal.

4.working process:

The working process of the Integrated Smart Alumni Networking and Career Support Portal starts when students, alumni, or administrators access the application through a web browser. Users can register or log in using their credentials. The frontend sends user requests to the backend, where the requests are processed according to the required functionality. The backend communicates with the MySQL database to store, retrieve, or update information. After processing the request, the backend sends the response to the frontend, and the result is displayed to the user. Through this process, students can search for alumni, send mentorship requests, explore job opportunities, and track their applications. Alumni can connect with students and provide career guidance, while administrators can manage users and monitor portal activities.

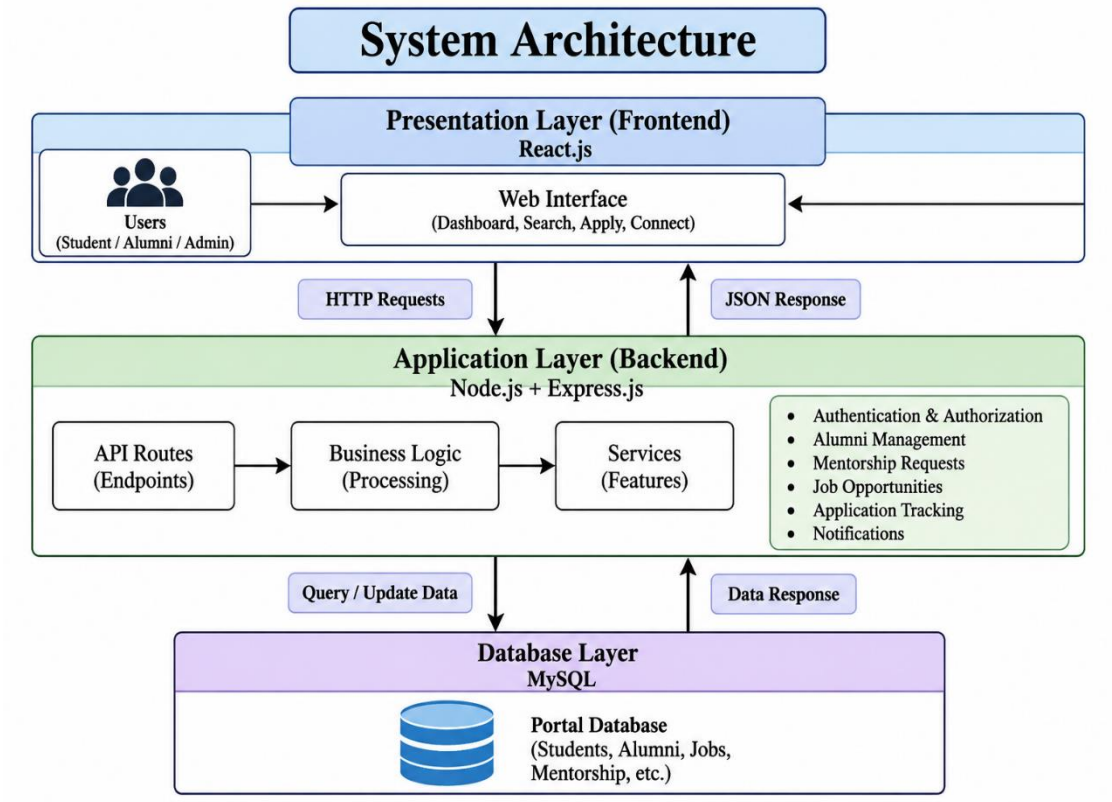


Fig 5.0 System Architecture

6.MODULES

The Integrated Smart Alumni Networking and Career Support Portal is divided into different modules to organize the system functionalities and make the application easier to manage. Each module performs a specific operation and helps students, alumni, and administrators interact with the portal.

1. User Registration and Login Module:

This module allows students and alumni to register using their personal details and access the portal through their login credentials. It also provides secure authentication

2. Forgot Password Module:

This module helps users recover their accounts when they forget their passwords by following the password recovery process.

3. Alumni Profile Management Module:

This module allows alumni to create, update, and manage their profiles, including educational qualifications, work experience, skills, and contact information.

4. Alumni Search Module:

This module helps students search for alumni based on their names, departments, graduation years, skills, and professional details.

5. Mentorship Request Module:

This module enables students to send mentorship requests to experienced alumni and receive career guidance and professional advice.

6. Job Opportunities Module:

This module allows alumni to share job openings and employment opportunities. Students can view available jobs and find suitable positions.

7. Job Application Tracking Module:

This module helps students track their job applications and check application progress and status.

8. Student Dashboard Module:

This module displays student information, alumni connections, mentorship requests, available jobs, and application updates.

9. Alumni Dashboard Module:

This module allows alumni to manage their profiles, view mentorship requests, and share job opportunities.

10. Notifications Module:

This module provides notifications about mentorship requests, job postings, application updates, and important portal activities.

11. Admin Management Module:

This module allows administrators to manage student and alumni accounts, monitor activities, and maintain portal information.

12. Search and Filter Module:

This module helps users find relevant alumni and job opportunities using different search options and filters.

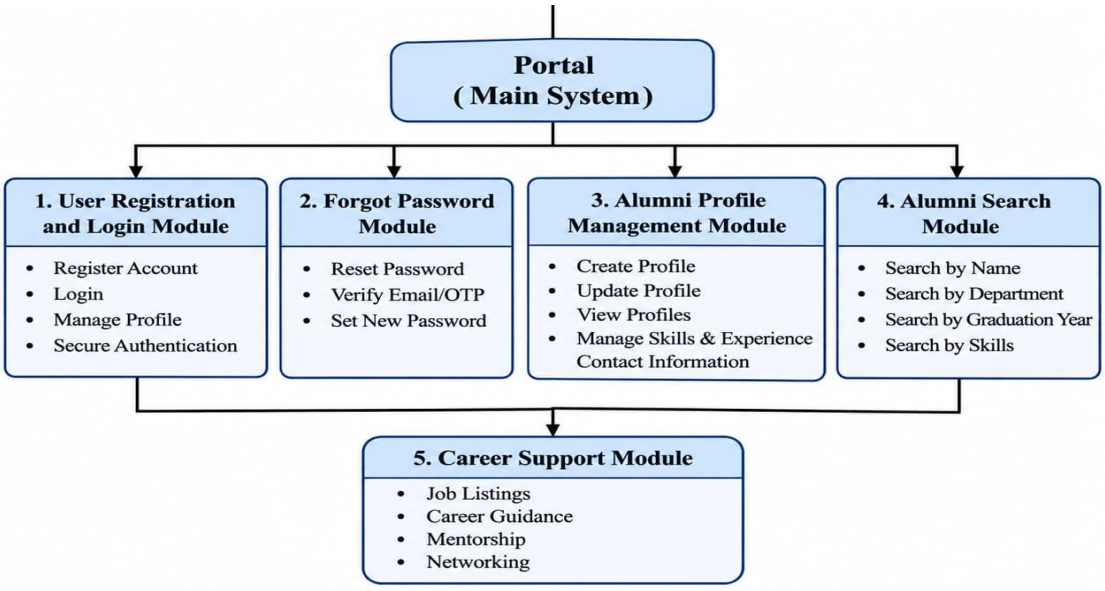


Fig 6.0 modules operations

7.UML DIAGRAM

The Integrated Smart Alumni Networking and Career Support Portal UML (Unified Modelling Language) is a standard visual language used to design, represent, and understand the structure and behaviour of a software application. It helps developers understand how different components of a system interact with each other. In our Integrated Smart Alumni Networking and Career Support Portal, UML diagrams are used to represent the interactions between students, alumni, and administrators. They explain the system's functionality, structure, and workflow in a clear and organized manner diagrams help in understanding the overall system design, identifying user activities, and explaining the relationships between different modules. They also make the development process easier by providing a visual representation of the application before implementation. UML provides different types of diagrams to represent various aspects of a software system. These diagrams explain how users interact with the system, how different components are connected, and how information flows between them. It also helps identify system requirements, reduce development complexity, and maintain proper documentation diagrams are mainly divided into two categories: Structural Diagrams and Behavioural Diagrams. Structural diagrams represent the static structure of the system, while behavioural diagrams describe the dynamic activities and interactions within the system. In the Integrated Smart Alumni Networking and Career Support Portal, UML diagrams are used to represent the overall system design, including student and alumni interactions, administrator activities, job opportunities, mentorship, and career support services. These diagrams provide a clear understanding of the system before implementation and help ensure efficient software development.

For the Integrated Smart Alumni Networking and Career Support Portal, the main UML diagrams are:

1. Use Case Diagram – Shows interactions between Student, Alumni, and Admin.
2. Class Diagram – Shows classes, attributes, and methods.
3. Sequence Diagram – Shows communication between users and system components.
4. Activity Diagram – Shows the workflow of the portal.
5. Component Diagram – Shows the major software components.
6. Deployment Diagram – Shows how the system is deployed on servers

7.1 class diagrams:

In our Integrated Smart Alumni Networking and Career Support Portal, the class diagram represents the main entities involved in the system, such as User, Student, Alumni, Admin, Job, Mentorship, and Application.Each class contains specific attributes and methods that define its properties and operations. The relationships between these classes explain how students interact with alumni, apply for jobs, request mentorship, and how administrators manage the portal.The class diagram helps developers understand the system structure, organize application components, and maintain a clear relationship between different modules.

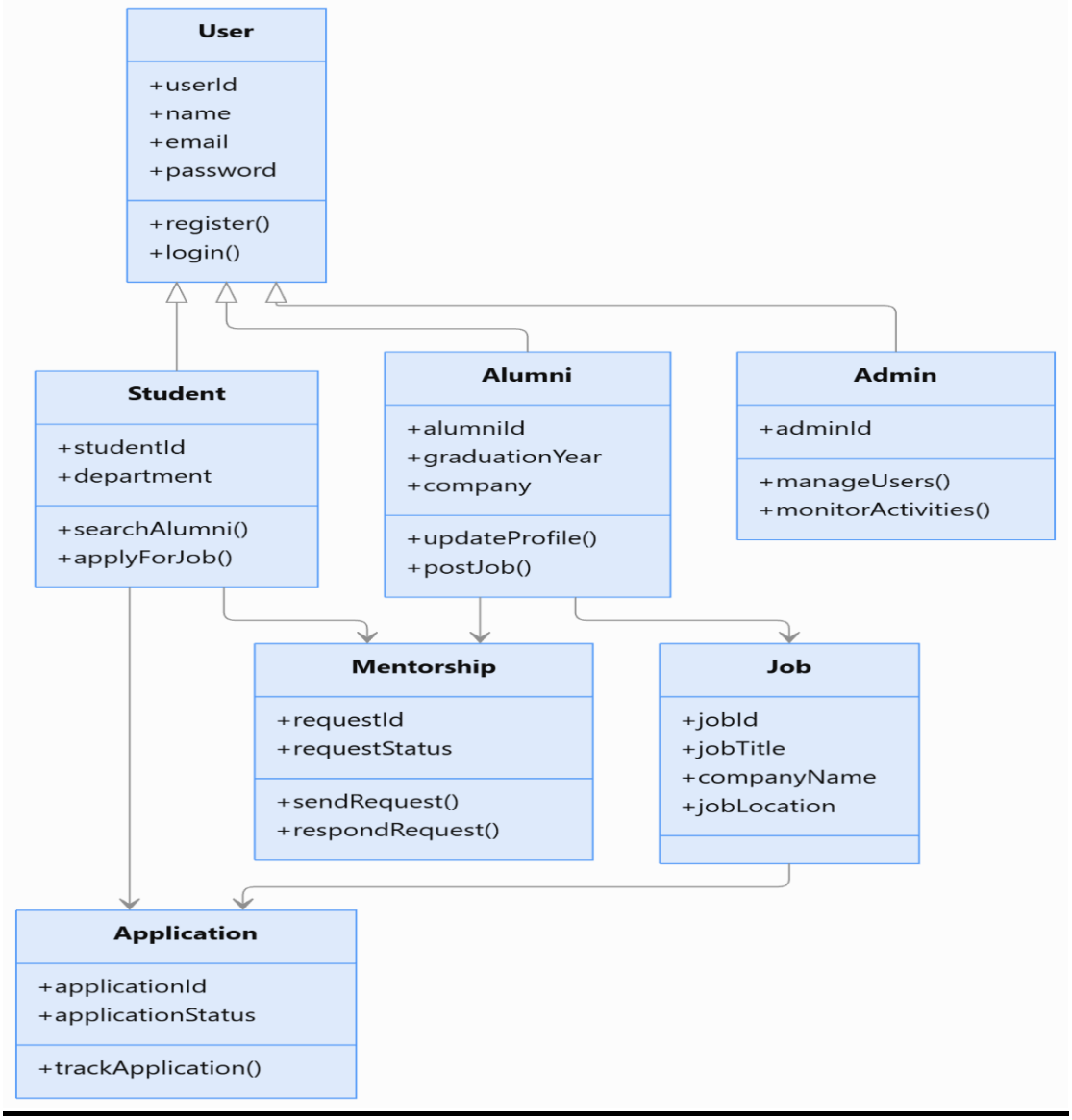


Fig 7.1 class diagrams

7.2 Use case diagram

Use Case Diagram represents three main actors: Student, Alumni, and Admin. Each actor performs different activities based on their role in the system.Students can register, log in, search for alumni, send mentorship requests, view job opportunities, apply for jobs, and track their applications. Alumni can manage their profiles, respond to mentorship requests, and post job opportunities. Administrators can manage user accounts, monitor activities, and maintain the portal.The Use Case Diagram helps to understand the system functionalities, identify user responsibilities, and explain the interaction between different users and the application.

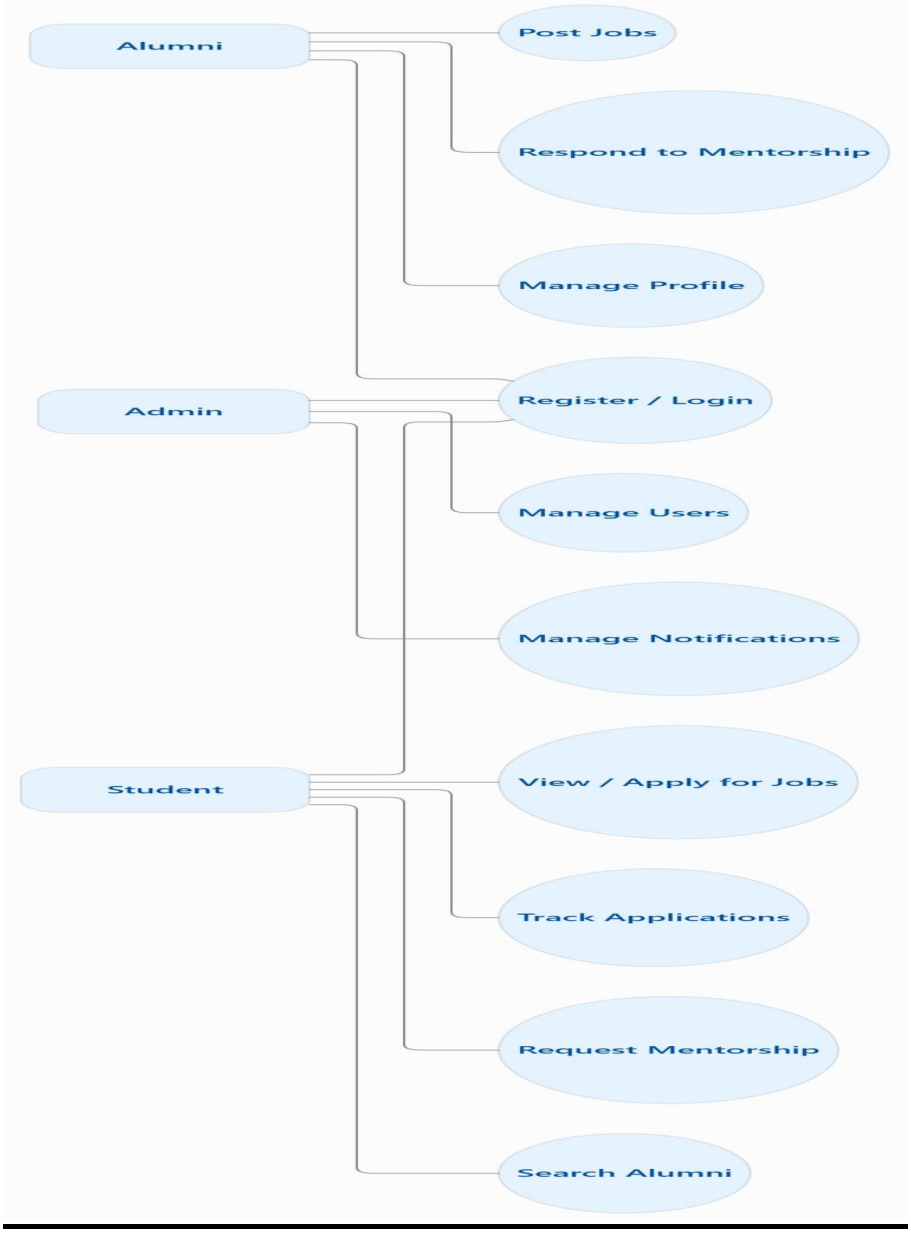


Fig 7.2 Use case diagram

7.3 Sequence Diagram

The Sequence Diagram explains how students, alumni, and administrators interact with the application. It represents the process of user registration, login authentication, alumni searching, mentorship requests, job applications, and database operations.When a user performs an action, the React.js frontend sends a request to the backend. The backend processes the request, communicates with the SQL database, and returns the response to the frontend. The result is then displayed to the user.The Sequence Diagram helps understand the order of operations, communication between components, and flow of information within the system.

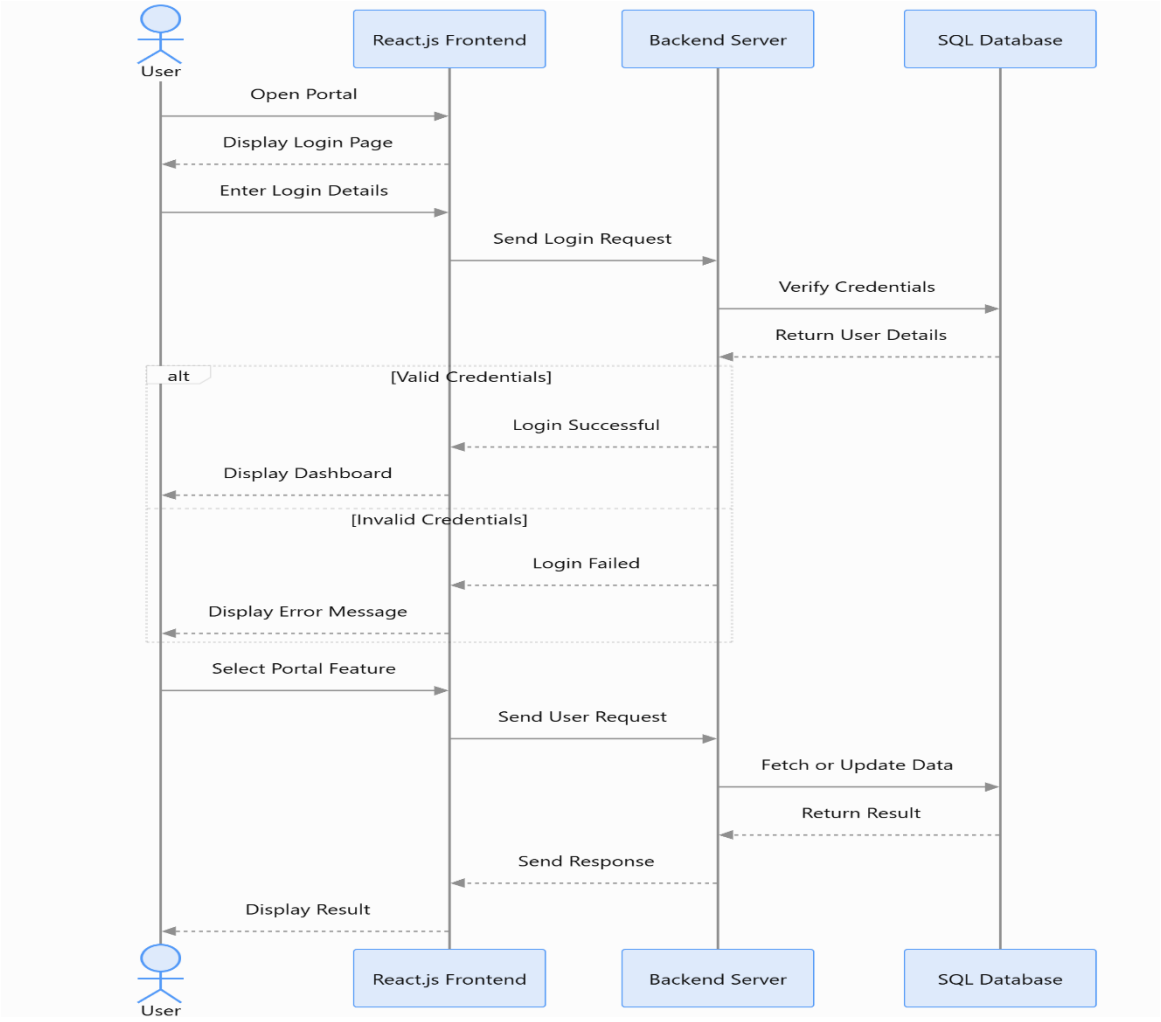


Fig 7.3 Sequence diagram

7.4 component Diagram

the Component Diagram represents the major components, including the React.js frontend, backend services, authentication module, alumni management, mentorship management, job management, notification system, and SQL database.The React.js frontend provides the user interface for students, alumni, and administrators. The backend processes user requests and manages application operations. The SQL database stores and retrieves user information, alumni profiles, mentorship requests, job details, and application records.The Component Diagram helps developers understand the system architecture, component relationships, and communication between different parts of the application

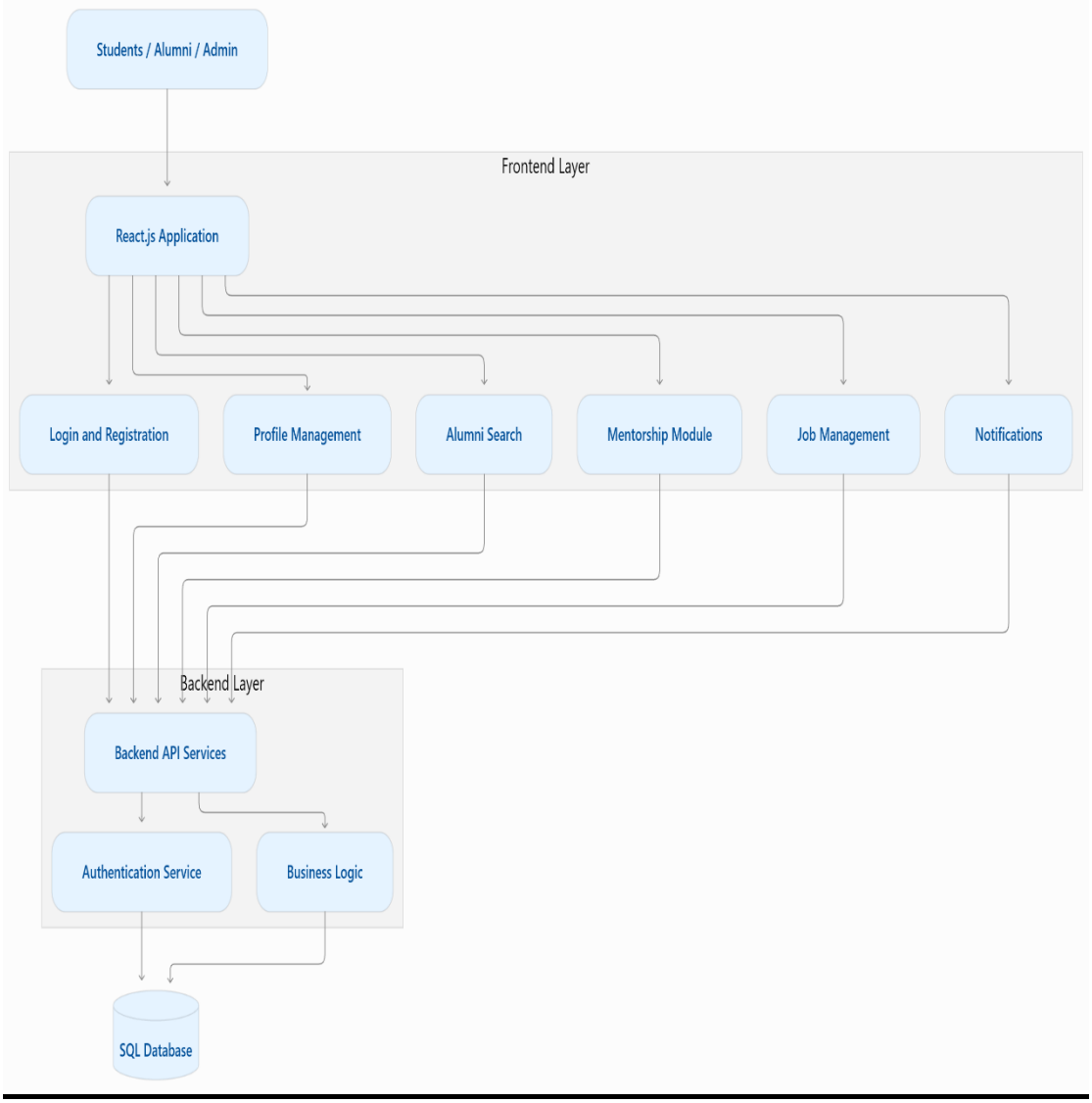


Fig 7.4 Component diagram

7.5 Deployment diagram:

A Deployment Diagram is a type of UML (Unified Modeling Language) diagram used to represent the physical arrangement of hardware and software components in a system. It shows how the application is deployed on different devices or servers and how these components communicate with each other.In our Integrated Smart Alumni Networking and Career Support Portal, the Deployment Diagram represents how

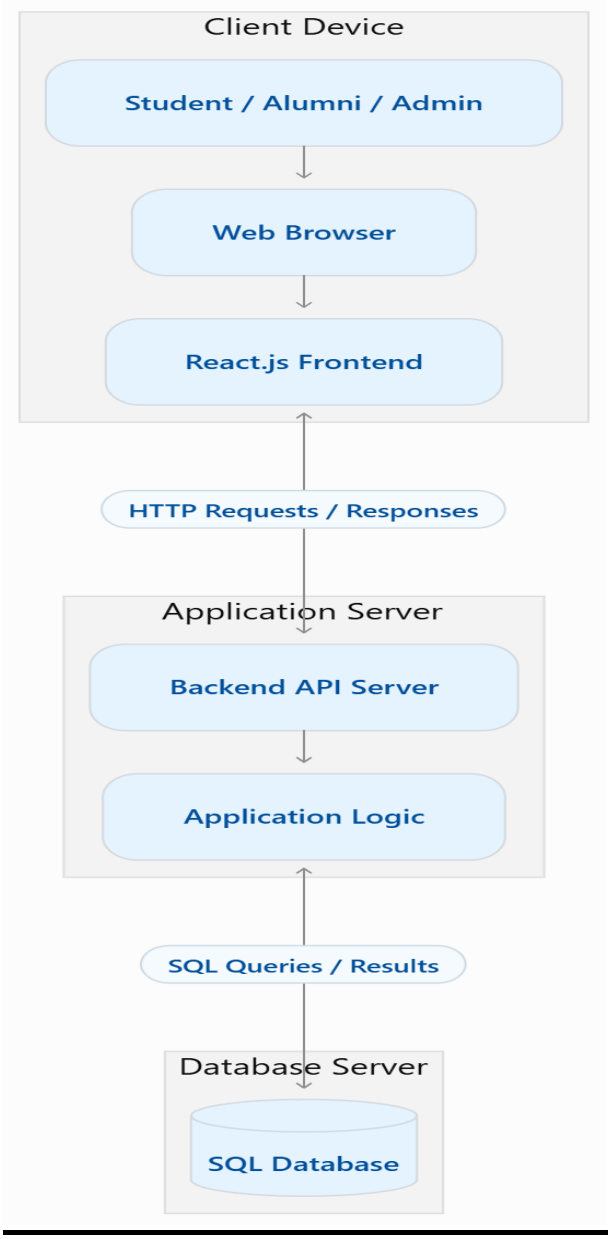


Fig 7.5 Deployment diagram

7.6 Activity Diagram

In our Integrated Smart Alumni Networking and Career Support Portal, the Activity Diagram explains the workflow starting from accessing the portal and logging in to performing different activities. It shows how students, alumni, and administrators interact with the system.Students can search for alumni, request mentorship, view job opportunities, and track applications. Alumni can manage their profiles, respond to mentorship requests, and post job opportunities. Administrators can manage users and monitor portal activities.

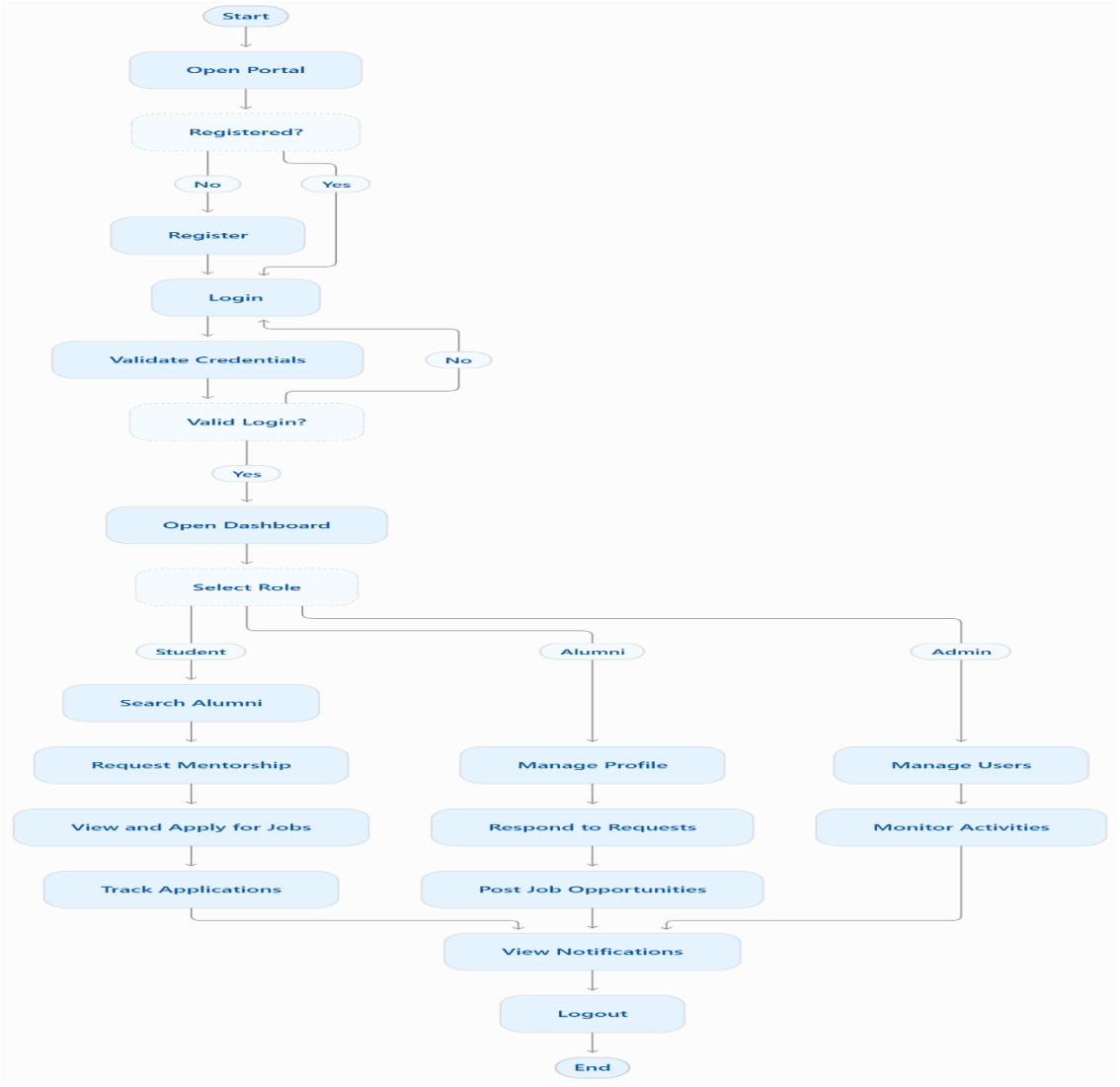


Fig 7.6 Activity diagram

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